

Namita Chougule

DATA ENGINEER

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SUMMARY

Data Engineer with 4 years of experience building and optimizing scalable data platforms, real-time streaming pipelines, and cloud-based data solutions across AWS environments. Hands-on expertise in Apache Spark, Kafka, Snowflake, Databricks, and Airflow, with a proven ability to develop batch and streaming ELT pipelines, improve query performance, and enhance production reliability. Experienced in processing high-volume datasets, optimizing multi-terabyte Spark workloads, and implementing data quality frameworks to support trusted analytics and machine learning initiatives. Strong foundation in distributed data systems, performance tuning, and delivering resilient, business-critical data solutions.

SKILLS

Programming Languages: Python, SQL, Bash Shell Scripting

Big Data & Distributed Processing: Apache Spark (PySpark), Apache Kafka, Delta Lake, Databricks, AWS EMR, StreamSets

Data Warehousing & Modeling: Snowflake, Dimensional Modeling (Star Schema), Data Warehouse Design, Query Optimization

Data Engineering & Pipeline Development: Batch & Streaming ELT/ETL Pipelines, Real-Time Data Processing, Data Ingestion Frameworks, Spark Performance Tuning

Workflow Orchestration & Data Quality: Apache Airflow, Great Expectations, Data Validation Frameworks, Pipeline Monitoring & Alerting

Cloud & Infrastructure: AWS (S3, EC2, EMR, VPC), Terraform, CI/CD

Core Competencies: Distributed Data Systems, Production Support & Reliability, Performance Optimization, Data Lake Architecture, Troubleshooting & Root Cause Analysis

EXPERIENCE

Quantum Integrators | July 2025 – Current | Data Engineer

- Developed reusable feature pipelines and supported feature store initiatives using Python and Databricks, improving data availability for downstream ML workloads by 25%.
- Improved analytical query performance by 35% for customer and operational dashboards by optimizing SQL transformations and implementing multi-layer Snowflake data models.
- Built batch and streaming ELT pipelines using StreamSets and Apache Kafka to ingest and consolidate over 1.5 million records daily from disparate on-premise systems into a centralized lakehouse environment.
- Enhanced production reliability for critical client data pipelines by implementing automated testing, monitoring, and GitOps-based deployment practices that supported 99.95% SLA objectives.

Infinite Infolab | Jan 2021 - Dec 2023 | Data Engineer

- Developed real-time ingestion pipelines using Apache Kafka and PySpark Structured Streaming on AWS EMR, enabling the platform to process peak volumes of 15,000 events per second and reducing data delivery latency from 6 hours to under 15 seconds.
- Contributed to the migration of legacy data pipelines to a Delta Lake architecture by implementing file optimization strategies and partitioning improvements, reducing storage utilization by 42% and improving query efficiency.
- Enhanced batch orchestration workflows using Apache Airflow by developing dynamic DAGs, custom sensors, and retry mechanisms that improved pipeline reliability and supported 99.9% SLA targets for critical reporting workloads.
- Designed dimensional data models and optimized Snowflake workloads through warehouse tuning and clustering strategies, reducing average query execution times by 68% and lowering compute consumption by 31%.
- Integrated Great Expectations into production pipelines to automate data validation checks and Slack-based alerting, improving the detection of data quality issues before downstream consumption.
- Developed Terraform modules to standardize provisioning of data platform resources across development, staging, and production environments, reducing deployment effort from days to under 20 minutes.
- Optimized multi-terabyte Spark workloads by replacing inefficient joins with broadcast strategies, tuning executor configurations, and mitigating data skew, reducing nightly processing times by 4.5 hours and eliminating recurring OOM failures.

EDUCATION

Master of Science in Computer Science

George Mason University, USA | GPA: 3.87/4.00

May 2025

Bachelor of Engineering in Computer Science & Engineering

Visvesvaraya Technological University, India | GPA: 8.18/10.00

Aug 2020

CERTIFICATIONS

AWS Certified Data Engineer – Associate

PROJECT

AI-Powered Hybrid Chatbot | Python, React, GPT, Pinecone, RAG

- Built a production-ready RAG system using GPT and Pinecone for contextual, document-aware responses.
- Developed FastAPI backend and React frontend for real-time conversational interactions.
- Implemented fallback retrieval mechanisms to improve response accuracy and system robustness.